

Memory Cycle at an Inflection Point

A Proprietary Cycle Framework for Micron, SK hynix and Samsung - Historical Analogues, Current Signal and Investment Implications

Investment conclusion

The latest monthly observation classifies Micron, SK hynix and Samsung Electronics in Stage 10 - Price-Led Roll-over. Forward EBITDA expectations are still being revised upward, but the rate of upward revision has slowed sharply and 3M stock returns have turned negative across all three names. This is not yet a confirmed earnings downturn. It is a late-cycle warning state in which price is beginning to discount a possible future earnings peak before consensus estimates fully roll over.

Report date: 10 August 2026

Latest cycle observation: August 2026 monthly snapshot

Coverage: Micron / SK hynix / Samsung Electronics

Frequency: Monthly | Historical benchmark: 2018-2023 | Current cycle: 2024-2026 held out from historical percentile calibration

Research framework: Rule-based quantitative cycle classifier using forward EBITDA revisions, revision acceleration, stock returns and valuation state

Independent Research Note

1. Executive Summary

The memory industry has moved beyond the cleanest part of the earnings-upgrade cycle. Our proprietary Memory Cycle Tracker indicates that Micron, SK hynix and Samsung Electronics have all transitioned into Stage 10 - Price-Led Roll-over. In this state, forward EBITDA revisions remain positive, but revision acceleration has turned sharply negative and equity prices have already weakened. The signal therefore sits between a normal high-cycle expansion and a confirmed earnings downturn.

The distinction matters because memory stocks often turn before reported fundamentals and before sell-side consensus estimates. Waiting for EBITDA revisions to become outright negative can be too late from a tactical perspective. At the same time, treating every early price decline as a confirmed peak is also dangerous: historical Stage 10 episodes include both genuine cycle tops and powerful re-acceleration phases. The current signal should therefore be interpreted as a decision point rather than an automatic exit signal.

Our historical framework also highlights a recurring valuation trap. In cyclical memory names, the lowest NTM EV/EBITDA multiples often occur when forward EBITDA is at or near a cycle high. A low multiple can therefore reflect a temporarily inflated denominator rather than a durable margin of safety. Conversely, very high multiples can appear near troughs when forward EBITDA has collapsed. For that reason, we treat valuation as a cycle-state confirmation variable rather than a standalone cheap-versus-expensive signal.

The most attractive historical risk-reward has generally appeared around the transition from deterioration to improvement: Trough Formation, Price-Led Bottom and Early Recovery. By contrast, High-Cycle Expansion has historically carried weaker forward returns because earnings expectations are already elevated and the market is vulnerable to a change in revision momentum. Stage 10 has shown particularly wide dispersion: short-term forward returns tend to be weak, while six-month outcomes depend heavily on whether earnings re-accelerate or roll over.

Bottom line for investors

We move from an expansion mindset to a confirmation mindset. Existing strategic positions do not need to be liquidated mechanically, but incremental risk-taking should be more selective. The cleanest bullish trigger would be a renewed turn upward in 3M Revision Acceleration together with improving stock momentum. The clearest bearish confirmation would be 3M EBITDA Revision crossing from positive to zero/negative while stock returns remain weak and the 36M EBITDA Peak Ratio begins to fall.

	A	B	C	D	E	F	G	H
1	Current Memory Cycle Snapshot 2026-08							
2	All metrics are derived from the user-built Memory Cycle Tracker; historical percentiles use the 2018-2023 company-specific benchmark.							
3								
4	Company	3M Stock Return	3M EBITDA Revision	3M Revision Accel	NTM EV/EBITDA	Multiple Hist Pctl	36M EBITDA Peak Ratio	Cycle Stage
5	Micron	(9.2%)	44.9%	(59.8%)	4.7x	38%	100%	Stage 10 - Price-Led Roll-over
6	SK hynix	(35.9%)	12.1%	(76.9%)	2.6x	23%	82%	Stage 10 - Price-Led Roll-over
7	Samsung Electronics	(27.3%)	19.7%	(81.1%)	2.2x	0%	100%	Stage 10 - Price-Led Roll-over

Figure 1. Current memory-cycle snapshot: all three companies are classified in Stage 10 - Price-Led Roll-over

Source: user-built Memory Cycle Tracker. Historical percentiles are benchmarked to each company's own 2018-2023 distribution.

2. Research Design and Data Architecture

The objective of the framework is not to forecast one quarter of revenue or EPS. It is to identify where the industry sits in the earnings and equity-price cycle, and to determine whether the current state historically resembled an attractive accumulation phase, a mature expansion, or an early deterioration phase. The framework therefore emphasizes forward-looking estimate changes and market response rather than trailing accounting ratios.

2.1 Core company set

The tracker uses Micron, SK hynix and Samsung Electronics as the core sample. Micron and SK hynix provide the cleanest read-through to DRAM/HBM and memory-cycle sensitivity. Samsung is more diversified, so its absolute valuation level should not be compared mechanically with the other two, but its direction of earnings revisions and stock-price response provides a useful breadth confirmation signal. A cycle signal that appears simultaneously in all three companies is more informative than a signal isolated to a single name.

2.2 Raw inputs

The model begins with three monthly series: cumulative stock performance, NTM EV/EBITDA and NTM EBITDA consensus. Stock performance is sourced from the Koyfin export used in the workbook. Historical EV/EBITDA and NTM EBITDA consensus values were digitized from Koyfin charts because the underlying estimate and valuation fields were not available for direct export. Accordingly, the framework is designed for directional and cycle-stage analysis rather than forensic point-in-time valuation to the second decimal place.

2.3 Why monthly frequency

Monthly data reduces noise relative to daily price moves and matches the slower-moving nature of analyst estimate revisions. A monthly horizon is also appropriate for a cycle model because the relevant question is not whether a stock reacts to one earnings print, but whether forward expectations are persistently improving or deteriorating over several months. Three-month transformations are used for the main signal to balance responsiveness and stability.

	A	B	C	D	E	F
4						
5	Variable	Calculation	What it measures	Cycle interpretation	Primary / Confirming	Benchmark
6	3M EBITDA Revision	$\text{NTM EBITDA}(t) / \text{NTM EBITDA}(t-3) - 1$	Direction of forward earnings expectations	+ = upgrades; - = cuts	Primary	Raw sign + 2018-2023 percentile
7	3M Revision Acceleration	$\text{Revisions3M}(t) - \text{Revision3M}(t-3)$	Whether upgrades/cuts are accelerating or decelerating	+ = improving momentum; - = slowing/worsening	Primary	2018-2023 percentile
8	3M Stock Return	$(1 + \text{Perf}(t)) / (1 + \text{Perf}(t-3)) - 1$	Market's leading price signal	Often turns before consensus EBITDA	Primary confirmation	2018-2023 percentile
9	EV/EBITDA Hist Percentile	Percentile of current NTM multiple vs 2018-2023	Where valuation sits in prior-cycle distribution	Low percentile often = high-cycle earnings; high percentile often = trough	Primary confirmation	Company-specific
10	Valuation Heat	$1 - \text{EV/EBITDA Hist Percentile}$	Transforms low multiple into high cycle-heat score	Higher = hotter part of cycle	Score input	0-100%
11	36M EBITDA Peak Ratio	$\text{NTM EBITDA} / \max(\text{NTM EBITDA over prior 36M})$	How close forward earnings are to recent peak	>90% = high/near-peak earnings; <70% = early/mid recovery or downturn	Primary	Rolling 36M
12	3M Multiple Change	$\text{EV/EBITDA}(t) / \text{EV/EBITDA}(t-3) - 1$	Whether multiple is expanding or compressing	Rising multiple during EBITDA cuts is often bearish	Confirming	Raw sign
13	Earnings - Price Gap	$3M \text{ EBITDA Revision} - 3M \text{ Stock Return}$	Whether fundamentals or price are moving faster	Negative = price ahead; positive = earnings ahead	Confirming	Raw spread
14	Cycle Heat Score	$35\% \text{ Revision Pct} + 25\% \text{ Accel Pct} + 20\% \text{ Valuation Heat} + 20\% \text{ Stock Pct}$	Single 0-100 cycle-temperature summary	Higher = hotter cycle, but cannot alone identify late-cycle deceleration	Summary only	2018-2023 percentiles

Figure 2. Core variables used in the Memory Cycle Tracker

Source: Cycle Rules sheet in the user-built workbook.

3. The Five Core Variables That Define the Cycle

The framework can be understood through five questions: Are forward earnings estimates rising or falling? Is the rate of change improving or deteriorating? Is the stock leading the estimates? How close is forward EBITDA to a recent cycle high? Where does the current valuation multiple sit relative to the company's own historical distribution? These questions correspond to the five variables below.

Variable	Calculation	What it measures	Cycle interpretation
3M EBITDA Revision	$\text{NTM EBITDA}(t) / \text{NTM EBITDA}(t-3) - 1$	Direction of forward earnings expectations	Positive = upgrades; negative = cuts. This is the primary direction variable.
3M Revision Acceleration	$\text{Revision3M}(t) - \text{Revision3M}(t-3)$	Change in revision momentum	Positive = improving/accelerating; negative = upgrades slowing or cuts worsening. This is the primary turning-point variable.
3M Stock Return	$P(t) / P(t-3) - 1$	Market-leading signal	Stocks often turn before consensus. Positive price with negative revision can flag a bottom; negative price with positive revision can flag a top.
36M EBITDA Peak Ratio	$\text{Current NTM EBITDA} / \text{rolling 36M max}$	Height of the earnings cycle	Below 70% suggests a low or rebuilding earnings base; above 90% suggests earnings are near a recent high.
EV/EBITDA Historical Percentile	Percentile rank vs 2018-2023 company history	Valuation state within the prior cycle	Low percentile often coincides with peak earnings; high percentile often coincides with trough earnings.

3.1 Supporting variables

Three supporting indicators help interpret the core state. First, 3M Multiple Change measures whether EV/EBITDA is expanding or compressing. Second, Earnings-Price Gap is defined as 3M EBITDA Revision minus 3M Stock Return; a negative value implies price is moving ahead of fundamentals, while a positive value implies earnings revisions are outpacing the stock. Third, Cycle Heat is a weighted summary of historical revision percentile, acceleration percentile, valuation heat and stock-return percentile. Cycle Heat is useful for measuring how hot the environment is, but it cannot by itself distinguish High-Cycle Expansion from Late-Cycle Deceleration.

3.2 Why revision acceleration matters more than the absolute EBITDA level near a top

At the top of a memory cycle, forward EBITDA can remain at a record level even after the marginal direction has started to weaken. A framework that uses only the EBITDA level will therefore react too late. Revision acceleration is designed to detect a change in slope. For example, forward EBITDA may still be revised upward by 20%, but if the prior three-month revision was +60%, the market is experiencing a material deceleration even though the headline estimate remains very strong. That is precisely the environment in which late-cycle equity behavior often begins.

Practical shorthand

Revision tells us the direction. Acceleration tells us the inflection. Stock return tells us whether the market is leading. Peak Ratio tells us how high the earnings level already is. EV/EBITDA percentile tells us where the valuation sits within the prior-cycle distribution.

4. A 12-Stage Memory Cycle Framework

The traditional memory cycle can be described in four broad blocks - downturn, trough/recovery, expansion/high cycle and late cycle/peak. For investment decisions, however, four states are too coarse. We therefore divide the cycle into 12 investable states. Stage 0 represents insufficient history and Stage 13 represents mixed or transitional signals; neither is treated as a core economic stage.

	A	B	C	D	E	F
17						
18	Stage	Name	Primary Rule	Typical Confirmers	Interpretation	Priority
19	0	Insufficient History	Missing 3M history	First months of sample	Do not classify	Highest
20	1	Downturn Acceleration	Rev<0; Accel<0; Stock<0; PeakRatio<70%	Multiple usually rising	Earnings cuts continue after earnings have moved well below peak	High
21	2	Deep Downturn / Capitulation	Rev<0; Rev Pctlt<-20%; Stock<0; Multiple Pctlt<=80%; PeakRatio<=65%	Weak price percentile	Deep earnings trough / capitulation-like state	Highest among downturn states
22	3	Trough Formation	Rev<0; Accel<0; Stock<0	Cuts becoming less severe	Bad news is still present but improving	High
23	4	Price-Led Bottom	Rev<0; Accel<0; Stock<0	Gap often negative	Price turns before consensus earnings	High
24	5	Early Recovery	Rev>0; Accel<0; PeakRatio<70%	Multiple compresses from high levels	Forward earnings have turned upward from a low base	Medium
25	6	Broad Recovery	Rev>0; Accel<0; not yet high-cycle	2/3+ memory names often positive	Normal recovery / expansion	Default positive state
26	7	Earnings Acceleration	Rev Pctlt<=80%; Accel<0; PeakRatio<=75%	Stock strong; multiple falls	Upgrades are historically strong and still accelerating	High
27	8	High-Cycle Expansion	Rev>0; Accel<0; PeakRatio<=90%; Multiple Pctlt<=30%	Low multiple / high earnings	High-profit stage; low multiple may be denominator-driven	High
28	9	Late-Cycle Deceleration	Rev>0; Accel<0; Stock<0; PeakRatio<=90%; Multiple Pctlt<=35%	Gap may turn negative	Earnings still high but upgrade momentum is fading	Very high
29	10	Price-Led Roll-over	Rev>0; Accel<0; Stock<0	Price leads consensus lower	Stock may be anticipating an earnings peak before analysts cut	Very high
30	11	Earnings Peak / Roll-over	Rev<0; Accel<0; Stock<0; PeakRatio<=90%	Multiple may start rising	Forward earnings peak is being confirmed	Very high
31	12	Downturn Confirmation	Rev<0; Accel<0; Stock<0; PeakRatio<=70%	Multiple often rises	Downcycle confirmed while earnings are still relatively elevated	Very high

Figure 3. Quantitative decision rules for the 12 core memory-cycle stages

Source: Cycle Rules sheet in the user-built workbook.

4.1 The cycle sequence

Stage	Name	Economic meaning
3	Trough Formation	Revisions are still negative, but acceleration turns positive; deterioration is slowing.
4	Price-Led Bottom	Revisions remain negative but improve, while the stock turns positive first.
5	Early Recovery	Revisions turn positive and accelerate from a still-low earnings base.
6	Broad Recovery	Positive revisions broaden and the earnings base normalizes.
7	Earnings Acceleration	Revision strength moves into a high historical percentile and remains accelerating.
8	High-Cycle Expansion	Forward EBITDA is near a 36M high and the valuation multiple sits at a low historical percentile.
9	Late-Cycle Deceleration	Earnings remain high and revisions remain positive, but acceleration turns negative.
10	Price-Led Roll-over	The stock turns negative before forward revisions turn negative.
11	Earnings Peak / Roll-over	Revisions stall or turn negative while earnings are still near peak.
12	Downturn Confirmation	Revisions, acceleration and stock momentum are negative while earnings remain relatively elevated.
1	Downturn Acceleration	Cuts continue as earnings move materially below the peak.
2	Deep Downturn / Capitulation	Revision and price weakness are extreme, earnings are far below peak, and the multiple often looks optically high.

4.2 Why the framework is intentionally rule-based

The current version is a quantitative rule-based classifier rather than a statistically optimized machine-learning model. Every row is classified from explicit numerical conditions, but thresholds such as a 90% Peak Ratio or a 30% multiple percentile were selected for economic interpretability rather than estimated from a large independent training sample. This is appropriate for the current dataset because the number of fully independent historical memory cycles is limited. The framework can later be calibrated through expanding-window cross-validation, but it should not be overfit to the current 2024-2026 cycle.

The key research discipline is therefore to keep 2018-2023 as the historical calibration period and treat 2024-2026 as an out-of-sample current-cycle observation. Rewriting the historical rules after seeing the current result would destroy the value of the test. The present framework is deliberately simple enough to remain interpretable and auditable.

What matters near a top

The most important late-cycle combination is Revision > 0, Acceleration < 0 and Stock Return < 0. This is Stage 10. It says consensus still looks good, but the market is beginning to price a future earnings slowdown.

5. Current Cycle Reading: Broad Stage 10 Confirmation

The latest monthly observation shows unusually broad confirmation. Micron, SK hynix and Samsung Electronics are all classified in Stage 10 - Price-Led Roll-over. The breadth matters because the three companies have different business mixes and absolute valuation levels. A simultaneous signal suggests the model is detecting an industry-level change in the slope of expectations rather than a company-specific event.

Company	Stock 3M	EBITDA Rev 3M	Accel 3M	NTM EV/EBITDA	Multiple Pctl	Peak Ratio	Stage
Micron	-9.2%	+44.9%	-59.8ppt	4.7x	38th	100%	Stage 10
SK hynix	-35.9%	+12.1%	-76.9ppt	2.6x	23rd	82%	Stage 10
Samsung	-27.3%	+19.7%	-81.1ppt	2.2x	0th	100%	Stage 10

5.1 Micron: earnings are still strong, but the slope has changed

Micron remains the strongest name on the headline earnings-revision measure, with 3M forward EBITDA revisions of approximately +44.9%. That number by itself looks highly bullish. The problem is the second derivative: 3M Revision Acceleration is approximately -59.8 percentage points and the stock's 3M return is about -9.2%. Forward EBITDA is still at 100% of its rolling 36M peak. The message is therefore not that Micron earnings are weak; it is that the market is no longer willing to extrapolate the prior pace of upgrades without evidence that the next leg of revisions can re-accelerate.

5.2 SK hynix: the most advanced cooling signal in the group

SK hynix shows the deepest price weakness, with a 3M stock return of roughly -35.9%, and revision acceleration of approximately -76.9 percentage points. Its 36M EBITDA Peak Ratio has already fallen to around 82%, below Micron and Samsung. This suggests that SK hynix is not only experiencing a slowdown in the pace of revisions; its forward earnings level has also moved further away from the recent peak. Among the three names, this is the clearest evidence of cooling.

5.3 Samsung: broad confirmation, but lower pure-memory beta

Samsung displays a similar directional signal: stock momentum is negative, revisions remain positive and acceleration is sharply negative. The absolute EV/EBITDA level is not directly comparable with Micron or SK hynix because Samsung is more diversified, but the consistency of the estimate and price pattern strengthens the industry read-through. From a portfolio perspective, Samsung may have lower pure-memory beta, but its cycle state still argues against assuming the memory up-cycle is in an early or middle phase.

Memory Cycle Relationship Dashboard									
Core interpretation: compare 3M stock return with 3M NTM EBITDA Revision. EV/EBITDA explains whether the gap is being driven by multiple expansion/compression.									
Company	3M Stock Return	3M EBITDA Revision	Revision Acceleration	NTM EV/EBITDA	3M Multiple Change	Earnings - Price Gap	Current Signal	Current Cycle Stage	Cycle Reason
Micron	(9.2%)	44.9%	21.9%	4.7x	(35.3%)	54.1%	Earnings-led expansion	Stage 10 - Price-Led Roll-over	Rev3M +44.9%, Accel3M -59.8ppt, Stock3M -9.2%, EV/EBITDA 4.7x (38th pctl); PeakRatio3M 100%; Multiple3M -35.3%; Earnings-Price Gap +54.1%; Cyclebeat 52%. The stock has turned negative before forward EBITDA revisions have turned negative; price may be leading a future earnings slowdown.
SK hynix	(35.9%)	12.1%	(36.7%)	2.6x	(41.9%)	48.1%	Earnings-led expansion	Stage 10 - Price-Led Roll-over	Rev3M +12.1%, Accel3M -76.9ppt, Stock3M -35.9%, EV/EBITDA 2.6x (23rd pctl); PeakRatio3M 82%; Multiple3M -41.9%; Earnings-Price Gap +48.1%; Cyclebeat -38%. The stock has turned negative before forward EBITDA revisions have turned negative; price may be leading a future earnings slowdown.
Samsung Electronics	(27.3%)	19.7%	7.5%	2.2x	(47.2%)	46.9%	Earnings-led expansion	Stage 10 - Price-Led Roll-over	Rev3M +19.7%, Accel3M -81.1ppt, Stock3M -27.3%, EV/EBITDA 2.2x (0th pctl); PeakRatio3M 100%; Multiple3M -47.2%; Earnings-Price Gap +46.9%; Cyclebeat 54%. The stock has turned negative before forward EBITDA revisions have turned negative; price may be leading a future earnings slowdown.

Figure 4. Workbook dashboard: current cycle stage and underlying reason by company

Source: user-built Memory Cycle Tracker.

6. How the Current Cycle Reached Stage 10

The current signal did not appear suddenly. The recent sequence across the three companies broadly followed the expected late-cycle progression: strong earnings acceleration, high-cycle expansion, deceleration and finally price-led roll-over. This sequence is important because it reduces the probability that the latest Stage 10 signal is merely an isolated monthly anomaly.

Company	Acceleration	High cycle	Deceleration	Latest
Micron	Stage 7 - Earnings Acceleration	Stage 8 - High-Cycle Expansion	Stage 9 - Late-Cycle Deceleration	Stage 10 - Price-Led Roll-over
SK hynix	Broad Recovery / Stage 7	Sustained Stage 7	Stage 9 in the latest transition	Stage 10
Samsung	Sustained Stage 7	Stage 8	Stage 9	Stage 10

6.1 The economic interpretation

In Stage 7, upward revisions are not only positive but historically strong and still accelerating. In Stage 8, the earnings base has reached a high level and the valuation multiple often looks optically low because EBITDA is elevated. Stage 9 begins when the same high earnings level is maintained but revision acceleration turns negative. Stage 10 occurs when price then weakens before the revision itself becomes negative. The market is effectively saying that current earnings strength is no longer sufficient evidence of future earnings durability.

6.2 Why this is more serious than a valuation correction alone

A pure valuation correction would typically involve price weakness without a material deterioration in the slope of earnings revisions. The current pattern is different: all three names show sharply negative revision acceleration. That does not prove a downturn, but it means the equity correction is occurring at the same time as the fundamental impulse is slowing. The combination deserves more weight than price weakness alone.

6.3 Why it is still too early to call a confirmed downturn

Despite the deterioration in momentum, the level of 3M EBITDA Revision remains positive across all three names. A confirmed Stage 11/12 transition would require revisions to stall or turn negative while stock momentum remains weak and the Peak Ratio begins to fall more broadly. In other words, the current state is late cycle with downside asymmetry, but not yet a completed earnings-peak confirmation.

Current state in one sentence

The market has moved from rewarding the level of earnings to questioning the durability of the earnings trajectory.

7. Historical Stage Behavior: Where Was the Best Risk-Reward?

To evaluate whether the stage labels carry investment meaning, we compare each monthly historical stage with subsequent 3M and 6M stock returns over the 2018-2023 benchmark period. This is descriptive rather than a formal independent backtest: monthly horizons overlap, sample counts differ by stage and only a limited number of fully independent memory cycles are available. The results should therefore be interpreted as a map of asymmetry rather than a precise expected-return forecast.

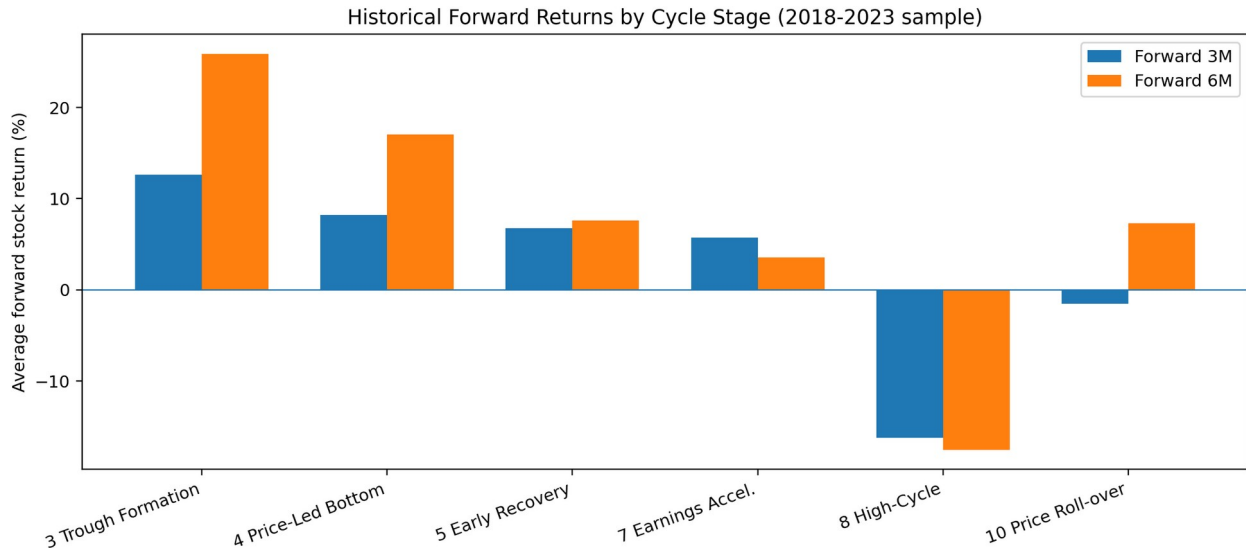


Figure 5. Historical average forward stock returns by cycle stage, 2018-2023 benchmark sample

Source: calculations from the user-built Memory Cycle Tracker. Pooled monthly observations; forward horizons overlap.

7.1 Trough and early-recovery states historically offered the cleanest asymmetry

The strongest historical pattern is that early improvement states have generally delivered better forward returns than mature high-cycle states. In the descriptive sample, Stage 3 - Trough Formation produced average forward returns of roughly +12.6% over 3M and +25.9% over 6M, while Stage 4 - Price-Led Bottom produced approximately +8.2% and +17.0%. These states occur when the level of earnings is still weak, but the direction of deterioration is improving and price may already be anticipating the turn.

7.2 High-Cycle Expansion historically looked cheap but carried weaker forward returns

Stage 8 - High-Cycle Expansion produced negative average forward returns in the historical sample, around -16.2% over 3M and -17.5% over 6M. The result illustrates the central valuation trap in memory stocks. A low EV/EBITDA multiple in a high-profit environment can create the appearance of cheapness precisely when the forward EBITDA denominator is most vulnerable to normalization. The correct question is not "Is the multiple low?" but "Is the current forward EBITDA level sustainable, and are revisions still accelerating?"

7.3 Stage 10 is not uniformly bearish - it is a high-dispersion branching state

Across all Stage 10 monthly observations, short-term forward performance was weak, while six-month outcomes were more mixed. The average future 3M return was around -1.5%, while the average future 6M return was around +7.3%. The improvement at six months should not be interpreted as a bullish Stage 10 signal. It reflects the fact that some Stage 10 episodes became true cycle tops and others re-accelerated sharply. Dispersion, not average return, is the defining feature of the stage.

Historical lesson

The most attractive entry point historically was not the moment when earnings and margins looked best. It was the moment when bad news stopped getting worse and price began to lead the recovery.

8. Stage 10 Case Study: Can the Cycle Re-Accelerate After Rolling Over?

Yes. A move from a high-cycle state into Stage 10 does not guarantee that the cycle is over. To reduce repeated counting of the same episode, consecutive Stage 10 months are de-duplicated and only the first month of each contiguous Stage 10 run is treated as an episode start. In the 2018-2023 benchmark sample, 12 Stage 10 episode starts had complete 3M forward data. The average 3M return was roughly -3.4%, the median was approximately -7.4%, and only about 25% of episodes were positive after three months. The short-term skew was therefore clearly weak.

At six months, however, the distribution widened substantially. Among 11 episodes with complete 6M data, the average return was around +10.0%, the median was only about +0.8%, and approximately 55% were positive. The large gap between average and median shows that a small number of powerful re-acceleration episodes pulled the average higher. Stage 10 is therefore better understood as a branching point that requires confirmation than as a standalone directional trade.

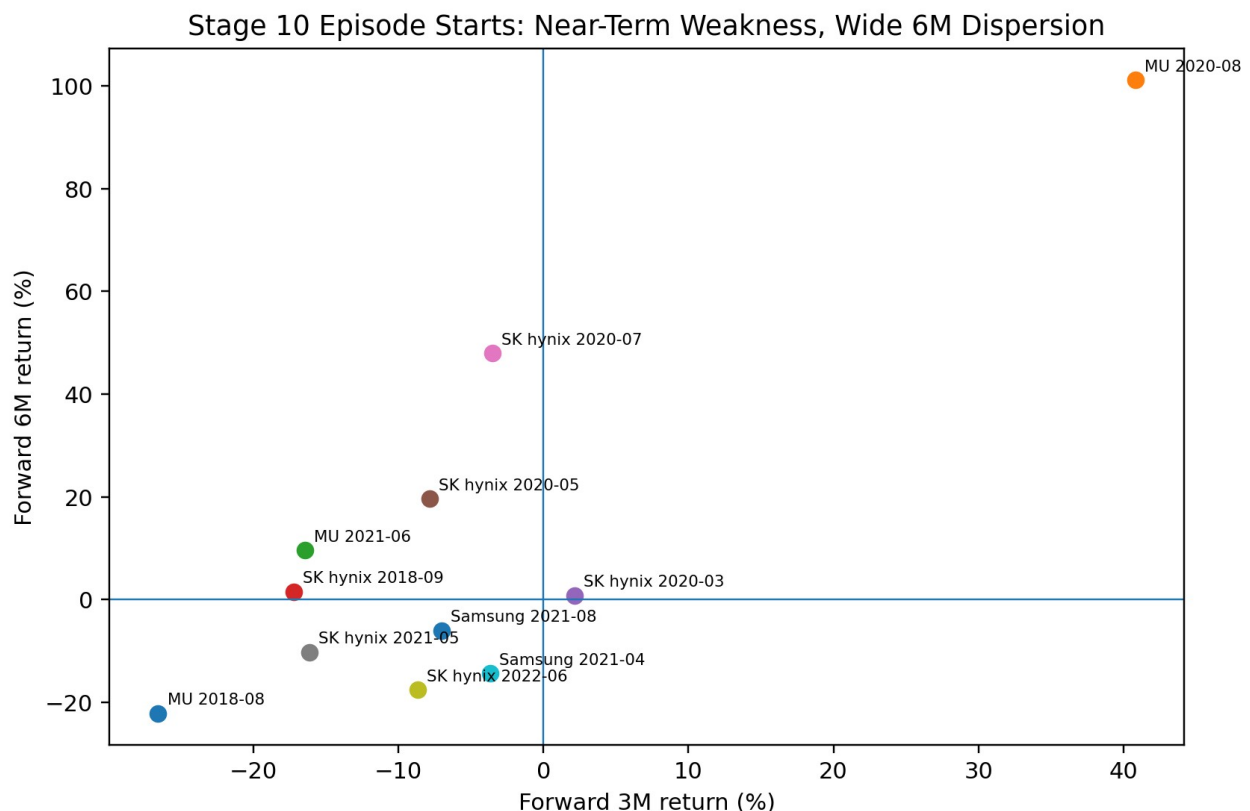


Figure 6. Stage 10 episode starts: weak short-term bias, highly dispersed six-month outcomes

Source: calculations from the user-built Memory Cycle Tracker. Consecutive Stage 10 months are de-duplicated to episode starts.

8.1 Historical examples

Episode	Starting state	Future 3M	Future 6M	Interpretation
Micron - Aug 2018	Stage 10	-26.6%	-22.2%	True cycle-top path. Price weakness was followed by persistent earnings deterioration and a traditional downcycle.
Micron - Aug 2020	Stage 10	+40.8%	+101.1%	Re-acceleration path. The early roll-over became a mid-cycle reset and revisions strengthened again.
SK hynix - Jul 2020	Stage 10	-3.5%	+47.9%	Short-term weakness, followed by a strong six-month recovery and renewed cycle strength.
SK hynix - Jun 2022	Stage 10	-8.7%	-17.6%	True downturn path. The price signal was followed by a broader earnings downcycle.

8.2 What separates re-acceleration from a true peak?

The most useful discriminator is the behavior of Revision Acceleration after Stage 10. In a re-acceleration path, acceleration improves from deeply negative toward zero and then positive while the 3M EBITDA Revision remains above zero. Price usually stabilizes or turns positive before consensus fully recovers. In a true peak path, acceleration remains negative and the revision itself eventually crosses below zero; the Peak Ratio then begins to fall and EV/EBITDA may rise even if the stock continues to decline because the denominator is falling faster than enterprise value.

This distinction is essential for the current cycle. A Stage 10 reading by itself should not trigger a mechanical bearish conclusion. What matters is the next transition. The model becomes materially more bearish if the industry moves into Stage 11 - Earnings Peak / Roll-over and then Stage 12 - Downturn Confirmation. Conversely, a return to Stage 6/7/8 would indicate that the current correction was a mid-cycle reset rather than the final top.

9. Valuation Implications: Why a Low EV/EBITDA Multiple Can Be a Late-Cycle Trap

Memory stocks are unusual because the denominator of EV/EBITDA is itself highly cyclical. When spot pricing, contract pricing, utilization and product mix improve, forward EBITDA can rise much faster than enterprise value. The multiple therefore compresses into what appears to be a cheap range. If investors anchor on that low multiple without asking whether forward EBITDA is near a peak, they can mistake cyclical earnings inflation for structural undervaluation.

The reverse occurs at troughs. Forward EBITDA can collapse faster than enterprise value, causing EV/EBITDA to rise sharply even when the stock has already fallen substantially. A high multiple near the bottom can therefore coincide with attractive future returns. This inversion is why the framework uses the multiple percentile only after identifying the direction and momentum of earnings revisions.

9.1 How to interpret the current multiples

The current NTM EV/EBITDA readings in the workbook are approximately 4.7x for Micron, 2.6x for SK hynix and 2.2x for Samsung. The absolute levels are not directly comparable across companies, particularly given Samsung's diversified business mix. More informative are each company's historical percentiles: Micron sits around the 38th percentile, SK hynix around the 23rd percentile and Samsung near the bottom of its historical distribution. These readings are consistent with a still-hot earnings environment, not necessarily with deep fundamental undervaluation.

9.2 The right valuation question in Stage 10

At this stage, investors should ask what multiple would be appropriate if forward EBITDA normalized, not what upside is implied by applying a mid-cycle multiple to peak earnings. A robust valuation process would therefore use scenario analysis: a traditional-cycle normalization case with lower forward EBITDA, a base case where current consensus is broadly sustained, an HBM structural-premium case where higher-quality AI-related demand extends the earnings duration, and an upside supercycle case where both earnings and the multiple remain above historical norms.

The current workbook is designed primarily for cycle-state identification rather than a precise target-price calculation, because the historical EBITDA and valuation series are digitized approximations. The appropriate next step for a formal target-price model would be to combine the stage framework with exact current net debt, diluted shares and a scenario range for NTM EBITDA and EV/EBITDA.

Valuation discipline

Do not buy a memory stock only because NTM EV/EBITDA looks low. First determine whether the low multiple reflects underpriced durable earnings or simply a temporarily elevated forward EBITDA denominator.

10. Forward Scenario Framework: The Next 1-3 Months Matter More Than the Current Level

The current cycle should be monitored as a transition problem. We define three practical forward paths. None requires forecasting an exact DRAM or HBM price. Instead, each path is identified by observable changes in revisions, acceleration and price behavior.

Scenario	Observable trigger	Likely stage path	Portfolio implication
Bull - Re-acceleration	3M Revision remains > 0; Acceleration rises from negative to positive; 3M Stock Return turns positive; Peak Ratio stabilizes or re-approaches highs	Stage 10 -> Stage 6/7/8	Treat the current drawdown as a mid-cycle reset. Add back cyclical beta selectively; Micron and SK hynix offer the greatest upside sensitivity.
Base - Late-cycle consolidation	Revision remains > 0; Acceleration remains < 0 but stops deteriorating; stock trades sideways/volatile	Stage 10 / Stage 13 transition	Maintain strategic core exposure but avoid chasing. Require confirmation before adding risk.
Bear - Peak confirmation	Revision moves toward 0 and turns negative; Acceleration remains < 0; stock remains weak; Peak Ratio falls more broadly	Stage 10 -> 11 -> 12	Reduce cyclical exposure, avoid averaging down on low multiples, and wait for Stage 3/4 bottom formation.

10.1 What would make the current view more bullish?

The clearest improvement would be a turn in 3M Revision Acceleration. The absolute revision level is already positive, so another month of strong EBITDA is not enough. We need evidence that the rate of upward estimate change is improving again. Ideally, this would be accompanied by a turn in 3M stock returns from negative to positive. That combination would indicate the market has begun to price a new earnings leg rather than merely reacting to a temporary valuation rebound.

10.2 What would make the current view materially more bearish?

The key bearish confirmation would be 3M EBITDA Revision crossing from positive to zero or negative while acceleration remains below zero. If the Peak Ratio simultaneously falls from near 100% toward lower levels and EV/EBITDA begins to rise despite continued share-price weakness, the pattern would closely resemble a traditional cyclical earnings roll-over. At that point, the low-multiple argument would become less persuasive because the denominator would be in active decline.

10.3 The role of HBM and AI demand

The largest reason the current cycle may differ from earlier memory cycles is the potential structural improvement in demand quality and product mix from HBM and AI infrastructure. If HBM meaningfully extends the duration of high utilization, pricing power and high-value memory demand, the industry could sustain elevated earnings longer than historical DRAM/NAND cycles suggest. The framework is deliberately designed to detect that possibility through re-acceleration rather than assume it ex ante. If the structural thesis is real, it should eventually appear in forward revisions and price behavior.

The single most important near-term monitor

3M Revision Acceleration. If it remains deeply negative, Stage 10 risk persists. If it turns back toward zero and then positive while revisions stay above zero, the probability of a renewed Stage 7/8 expansion rises materially.

11. Investment Strategy: Shift From "Chasing Earnings" to "Waiting for Confirmation"

11.1 Sector stance: Neutral to Cautious

The industry-level stance is Neutral to Cautious. This is not a call for indiscriminate short exposure, nor is it an argument that the long-term AI/HBM thesis is broken. It reflects a change in tactical asymmetry. The prior stage rewarded participation in upward estimate revisions. The current stage requires greater selectivity because the earnings impulse is slowing and price has already turned down. The expected value of adding exposure is therefore lower unless the next data points show re-acceleration.

For existing strategic positions, a partial core holding remains reasonable because Stage 10 can re-accelerate. However, position sizing should recognize that downside can become nonlinear if the revision itself turns negative. Investors who entered during earlier recovery stages have more flexibility to hold through volatility; investors considering new exposure should demand clearer evidence that the current correction is a reset rather than the first leg of a full downcycle.

11.2 Company-level positioning

Company	View	Why	Action
Micron	Neutral / Tactical Hold	Headline revisions remain strong (+44.9%), but acceleration is deeply negative and stock momentum has turned down. Highest upside beta if revisions re-accelerate, but also direct downside sensitivity if Stage 11/12 emerges.	Do not chase. Add only on improving acceleration/price confirmation or after a more complete cycle reset.
SK hynix	Cautious	Deepest stock drawdown, most negative acceleration and Peak Ratio already down to ~82%. Current cooling is further advanced than peers.	Keep tactical exposure lighter until forward earnings re-accelerate or a new bottoming state forms.
Samsung	Neutral / Relative Defensive	Similar late-cycle direction, but diversified earnings base lowers pure memory beta. Valuation percentile is very low, yet the cycle signal argues against treating this alone as a buy signal.	More defensive way to retain sector exposure, but still wait for cycle confirmation before increasing memory-specific risk.

11.3 A practical position-sizing framework by stage

Stage	Regime	Suggested posture	Rationale
Stage 3-4	Trough Formation / Price-Led Bottom	Increase risk budget	Historically attractive asymmetry; deterioration is improving and price may be leading the turn.
Stage 5-7	Early Recovery / Earnings Acceleration	Hold or add selectively	Fundamental direction and momentum are aligned.
Stage 8	High-Cycle Expansion	Hold, reduce valuation complacency	Earnings are strongest, but low multiples can be denominator-driven and future asymmetry worsens.
Stage 9-10	Late-Cycle / Price-Led Roll-over	Stop chasing; preserve optionality	Wait for re-acceleration or confirmed deterioration.
Stage 11-12	Earnings Peak / Downturn Confirmation	Reduce cyclical beta	Do not average down only because the multiple looks cheap.
Stage 1-2	Downturn / Deep Downturn	Prepare for next bottom	Watch for acceleration improving from negative territory and price stabilization.

11.4 What not to do

- Do not equate a low NTM EV/EBITDA multiple with low risk when forward EBITDA is at or near a cycle high.
- Do not wait for reported earnings to collapse before recognizing a top; the equity market and revision acceleration can turn first.

- Do not treat Stage 10 as an automatic sell signal; some historical Stage 10 episodes re-accelerated sharply.
- Do not change the historical thresholds to fit the current cycle after observing the outcome; preserve the out-of-sample discipline.

12. Monitoring Checklist for the Next Three Monthly Observations

The value of a cycle model comes from repeated, disciplined updates. The next one to three monthly observations should be evaluated in the same order every time. The following checklist is designed to prevent narrative drift and avoid overreacting to one headline number.

Priority	Metric	Question	Interpretation
1	3M Revision Acceleration	Is it still deeply negative, stabilizing, or turning positive?	Most important early indicator of re-acceleration versus peak confirmation.
2	3M EBITDA Revision	Does it remain > 0, approach 0, or turn negative?	Crossing below zero materially increases the probability of Stage 11/12.
3	3M Stock Return	Does price stabilize and turn positive before consensus?	Positive price with improving acceleration supports a reset/re-acceleration thesis.
4	36M EBITDA Peak Ratio	Does it remain near 100% or fall persistently?	A persistent decline confirms that forward earnings are moving away from the peak.
5	EV/EBITDA percentile and 3M change	Does the multiple rise as the stock falls?	Can indicate denominator compression as forward EBITDA is cut.
6	Industry breadth	Are 2/3 or 3/3 companies moving in the same direction?	Broad confirmation reduces the probability of a company-specific false signal.

12.1 Bullish transition checklist

- Revision remains positive.
- Acceleration improves for at least two consecutive monthly observations and ideally turns positive.
- Stock 3M return turns positive or materially outperforms while revisions stabilize.
- Peak Ratio stabilizes and does not continue to fall across the industry.
- The stage classification moves from 10 toward 13/6/7 rather than 11/12.

12.2 Bearish transition checklist

- Revision moves toward zero and becomes negative.
- Acceleration remains negative or deteriorates further.
- Stock returns remain negative across at least two of the three companies.
- Peak Ratio declines and forward EBITDA no longer revisits prior highs.
- EV/EBITDA begins to expand while the stock falls, indicating the denominator is compressing faster than enterprise value.

Research discipline

A single strong earnings print should not override the state model. The question is whether the forward revision path changes, not whether the latest reported quarter was strong.

13. Methodological Limitations and Next Research Steps

The framework is intentionally transparent, but it has limitations. First, the current stage thresholds are economically motivated rules rather than statistically optimized hyperparameters. A more advanced version should perform expanding-window or cycle-aware cross-validation on the 2018-2023 development period, then freeze the rules before evaluating the 2024-2026 current cycle.

Second, the number of independent full memory cycles in the available history is limited. Monthly observations are highly overlapping, especially when evaluating 3M and 6M forward returns. Standard significance tests would therefore overstate the amount of independent information unless serial correlation is addressed. Any future regression work using overlapping return horizons should use HAC/Newey-West standard errors or non-overlapping validation windows.

Third, historical EV/EBITDA and NTM EBITDA consensus values are digitized approximations from Koyfin charts. This is sufficient for broad cycle-state work but not ideal for precise point-in-time fair-value estimation. A formal valuation model should use exact historical consensus snapshots where available, together with reported cash, debt and diluted share counts.

Fourth, the present cycle may contain a structural regime shift because of HBM and AI infrastructure demand. A traditional memory-cycle framework can underestimate the duration of high earnings if product mix, supply discipline or customer demand quality has improved structurally. The correct response is not to discard the historical framework, but to test whether the current cycle repeatedly generates out-of-sample deviations that are consistent with a durable structural premium.

13.1 Recommended next quantitative experiments

- Fair-Multiple Model: regress $\log(\text{NTM EV/EBITDA})$ on 3M EBITDA Revision and Revision Acceleration using 2018-2023 development data, then test 2024-2026 out of sample.
- Future-Return Model: regress Future 3M Stock Return on Revision, Acceleration and current NTM EV/EBITDA, using a three-month purge at validation boundaries to avoid target leakage.
- Lead-Lag Test: measure whether stock returns lead future EBITDA revisions by one, two or three months. This directly tests the Stage 10 intuition that price may anticipate consensus.
- Stage Threshold Calibration: treat Peak Ratio and percentile cutoffs as hyperparameters and select them only within historical expanding-window validation, never using the current cycle for tuning.
- Panel Robustness: after running company-level models separately, pool the three companies with company fixed effects to test whether the relationships are industry-wide.

What would make the framework materially stronger

More independent historical cycles, exact point-in-time consensus data, explicit HBM/DRAM/NAND pricing variables, and out-of-sample validation of the current cycle. The current framework should be viewed as an auditable decision system that can be improved incrementally rather than a black-box forecast model.

14. Conclusion

The memory industry is no longer in the cleanest part of the up-cycle. The latest data show a synchronized Stage 10 - Price-Led Roll-over signal across Micron, SK hynix and Samsung. Forward EBITDA revisions are still positive, which prevents us from declaring a confirmed earnings downturn, but revision acceleration has deteriorated sharply and stock prices have already turned lower. The market is therefore beginning to discount the possibility that the current earnings run-rate is less durable than the headline consensus numbers imply.

Historical comparisons argue against two simplistic conclusions. The first is that low EV/EBITDA means the stocks are automatically cheap; in memory, a low multiple often appears precisely when forward EBITDA is near a peak. The second is that Stage 10 guarantees a major downcycle. Some Stage 10 episodes, including historical Micron and SK hynix cases, re-accelerated strongly after a temporary reset. The correct interpretation is that Stage 10 is a branching point with weak near-term asymmetry and unusually high medium-term dispersion.

For investors, this changes the optimal behavior. Earlier in the cycle, the right trade was to participate in upward earnings revisions. At the current stage, the better trade is to preserve optionality and wait for confirmation. A bullish confirmation requires revision acceleration to improve and price to stabilize before forward revisions turn negative. A bearish confirmation requires the revision itself to cross below zero while the Peak Ratio declines and stock weakness persists.

Accordingly, our sector stance is Neutral to Cautious. We would not chase the group simply because headline earnings are strong or valuation multiples look low. We would retain strategic exposure selectively, favor disciplined position sizing, and use the next one to three monthly revision updates to determine whether the cycle is re-accelerating or transitioning into a true earnings downturn. The most attractive next major opportunity may come either from a verified re-acceleration back into Stage 7/8 or from a later Stage 3/4 bottoming setup after a fuller reset.

Final investment message

The current debate is not whether memory earnings are high - they are. The debate is whether the slope of forward earnings can turn higher again before consensus itself rolls over. Until that question is answered, the rational posture is selective exposure, tighter entry discipline and close monitoring of revision acceleration.

Appendix A. Data and Classification Notes

The latest company-level cycle data and classifications are taken from `memory_cycle_with_cycle_stages_no_charts.xlsx`. Company-specific historical percentiles use January 2018 through December 2023 as the benchmark distribution. The 2024-2026 current cycle is therefore observed relative to the prior-cycle distribution rather than used to define what constitutes a high or low percentile.

Future 3M and 6M stock returns used in the historical stage analysis are calculated from monthly cumulative stock performance. Stage 10 episode analysis de-duplicates contiguous Stage 10 months by using only the first month of each run, which reduces repeated counting of the same underlying roll-over episode. Because return horizons overlap and the sample is limited, the results are descriptive and should not be interpreted as an independently validated trading strategy.

Appendix B. Classification Logic in Compact Form

Key rule	Stage
Rev < 0, Accel > 0, Stock <= 0	Stage 3 - Trough Formation
Rev < 0, Accel > 0, Stock > 0	Stage 4 - Price-Led Bottom
Rev > 0, Accel > 0, low Peak Ratio	Stage 5 - Early Recovery
Rev strong, Accel > 0, Peak Ratio rising	Stage 7 - Earnings Acceleration
Rev > 0, Accel >= 0, Peak Ratio >= 90%, low multiple percentile	Stage 8 - High-Cycle Expansion
Rev > 0, Accel < 0, Stock >= 0, high Peak Ratio	Stage 9 - Late-Cycle Deceleration
Rev > 0, Accel < 0, Stock < 0	Stage 10 - Price-Led Roll-over
Rev <= 0, Accel < 0, Stock < 0, high Peak Ratio	Stage 11 - Earnings Peak / Roll-over
Rev < 0, Accel <= 0, Stock < 0, still-elevated earnings	Stage 12 - Downturn Confirmation
Rev < 0, Accel < 0, Stock < 0, Peak Ratio lower	Stage 1 - Downturn Acceleration
Extreme negative revisions + weak stock + high multiple percentile + low Peak Ratio	Stage 2 - Deep Downturn / Capitulation